

MI-21: Resolution

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope.](#)

1 MI-21: a rational counterexample to the geometric-mean norm conjecture

For positive definite matrices, write

$$A\#_t B = A^{1/2}(A^{-1/2}BA^{-1/2})^t A^{1/2}, \quad A\#B = A\#_{1/2}B.$$

With $A = \sum_i A_i$ and $B = \sum_i B_i$, the target asserts

$$\left\| \sum_i (A_i^s \#_t B_i^s)^r \right\| \leq \left\| \left(A^{(1-t)srp/2} B^{tsrp} A^{(1-t)srp/2} \right)^{1/p} \right\| \quad (1)$$

for every unitarily invariant norm, $0 \leq t \leq 1$, and $p, r, s > 0$ with $sr \geq 1$ [1, Conjecture 3].

Theorem 1.1. *Inequality (1) is false already for two summands of order two, the operator norm, $s = t = 1/2$, and $r = 2$. The same counterexample works for every $p > 0$ and has rational positive definite inputs satisfying $A = B = I_2$.*

Proof. Define the rational matrices

$$C = \text{diag} \left(\frac{12}{37}, \frac{21}{29} \right), \quad E = \text{diag} \left(\frac{35}{37}, \frac{20}{29} \right), \quad S = \frac{1}{17} \begin{pmatrix} 15 & 8 \\ 8 & -15 \end{pmatrix}.$$

They satisfy $C, E > 0$, $C^2 + E^2 = I_2$, and $S = S^* = S^{-1}$. Set

$$A_1 = C^2, \quad A_2 = E^2, \quad B_1 = SE^2S, \quad B_2 = SC^2S.$$

All four inputs are rational and positive definite. Also $A_1 + A_2 = B_1 + B_2 = I_2$. Thus the matrix inside the norm on the right of (1) is exactly I_2 , for every $p > 0$.

Put $D = SES$ and $G = C\#D$. Symmetry and unitary covariance of the geometric mean imply

$$E\#(SCS) = SGS.$$

For the chosen parameters, the left-hand matrix is therefore

$$L = G^2 + SG^2S.$$

We compute it exactly. For any 2×2 positive definite C, D , if

$$\delta = \sqrt{\det D / \det C}, \quad h = \text{tr}(C^{-1}D) + 2\delta,$$

then

$$C\#D = \frac{D + \delta C}{\sqrt{h}}, \quad (C\#D)^2 = \frac{(D + \delta C)^2}{h}. \quad (2)$$

Indeed, applying the Cayley–Hamilton identity to $C^{-1/2}DC^{-1/2}$ shows that its positive square root is

$$\frac{C^{-1/2}DC^{-1/2} + \delta I}{\sqrt{\text{tr}(C^{-1}D) + 2\delta}}.$$

Congruence by $C^{1/2}$ proves (2).

In the present example,

$$\delta = \frac{5}{3}, \quad h = \frac{61697295}{8682716}, \quad D + \delta C = \frac{1}{310097} \begin{pmatrix} 443355 & 33000 \\ 33000 & 605715 \end{pmatrix}.$$

Rational matrix multiplication in (2) gives

$$G^2 = \frac{1}{12158163} \begin{pmatrix} 3516940 & 616000 \\ 616000 & 6547660 \end{pmatrix},$$

and hence

$$L = \frac{1}{12158163} \begin{pmatrix} 8216600 & -985600 \\ -985600 & 11912600 \end{pmatrix}. \quad (3)$$

For $w = (1, -4)^T$, direct multiplication yields

$$Lw = \frac{1351000}{1350907}w, \quad \frac{1351000}{1350907} = 1 + \frac{93}{1350907} > 1.$$

Therefore $\|L\|_\infty > 1 = \|I_2\|_\infty$, contradicting (1). $\square$

Remark 1.2 (Exact certificate and scope). The counterexample is not based on a rounded numerical eigenvalue. Every displayed matrix and every comparison after (2) is rational. Its other eigenvalue is $7970200/12158163 < 1$. The parameter condition is met with $sr = 1$. No claim is made here about the restricted parameter regimes already established in the cited paper.

References

- [1] Shaima’a Freewan and Mostafa Hayajneh. On a conjecture related to the geometric mean and norm inequalities. *Mathematical Inequalities & Applications*, 27(1):193–200, 2024. Conjecture 3, page 195. URL: <https://files.ele-math.com/articles/mia-27-15.pdf>, doi:10.7153/mia-2024-27-15.